

PARSONS TRI-CITY, KS, USA

WMO: 745431

Lat: 37.328N

Lon: 95.504W

Elev: 265

StdP: 98.18

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.4	-18.2	0.8	-10.9	-15.8	1.0	-8.9	11.3	5.2	10.1	5.6	2.5	0	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	37.2	24.0	35.2	24.5	33.6	24.2	26.4	32.8	25.7	32.1	25.1	31.5	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.2	30.1	23.9	19.3	29.3	23.0	18.4	28.4	83.6	32.9	80.6	32.4	77.9	31.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	9.8	8.5	7.6	DB	-18.4	38.7	4.2	2.5	-21.5	40.5	-23.9	41.9	-26.2	43.4	-29.3	45.2
(5)				WB	-18.3	27.5	4.6	0.8	-21.6	28.1	-24.3	28.5	-26.9	29.0	-30.2	29.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	1.2	3.4	9.0	14.4	19.5	24.5	26.9	26.3	22.1	15.2	8.5	2.6
	DBStd	10.30	6.03	6.25	6.00	5.01	4.25	3.09	2.95	3.19	4.14	5.12	5.53	5.82
	HDD10.0	926	277	198	92	17	1	0	0	0	12	92	237	
	HDD18.3	2342	530	417	294	136	39	2	0	1	13	124	298	487
	CDD10.0	2577	6	14	59	149	295	433	524	507	362	174	46	8
	CDD18.3	949	0	0	4	18	75	185	266	248	124	27	2	0
	CDH23.3	10023	0	3	31	156	642	1886	3045	2774	1234	238	13	0
	CDH26.7	4536	0	0	4	26	179	779	1541	1429	519	58	1	0
Wind	WSAvg	3.0	3.4	3.5	3.8	3.8	3.0	2.8	2.3	2.1	2.3	2.8	3.3	3.2
Precipitation	PrecAvg	1058	36	39	72	108	153	151	98	87	106	88	67	53
	PrecMax	1486	108	117	171	299	270	459	232	198	247	235	186	142
	PrecMin	622	2	0	11	17	61	41	22	4	23	7	0	6
	PrecStd	193	24	31	46	60	64	90	51	44	53	55	50	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	23.1	26.5	29.4	32.2	35.9	39.4	39.7	36.9	30.8	25.1	20.5
		MCWB	12.3	14.4	17.5	19.6	22.2	25.0	23.5	24.1	22.6	21.7	16.7	15.9
	2%	DB	16.8	19.6	23.4	27.2	30.4	33.7	37.2	37.4	33.3	28.2	22.4	17.3
		MCWB	11.2	13.2	16.0	18.3	22.0	24.4	24.5	23.8	23.5	19.7	15.3	12.9
	5%	DB	13.9	17.0	21.2	25.1	28.7	32.4	35.1	35.4	32.0	26.1	20.0	14.2
		MCWB	8.6	11.7	14.2	17.4	21.5	24.2	24.9	24.2	23.0	18.3	14.6	10.2
	10%	DB	11.0	13.8	19.0	23.0	27.2	31.0	33.4	33.5	29.8	23.8	17.6	12.1
		MCWB	6.7	9.0	13.6	16.6	20.9	23.5	24.8	23.8	21.9	17.9	12.8	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.1	16.8	19.0	21.4	24.5	26.7	27.5	27.3	25.6	23.2	18.9	17.0
		MCDB	17.4	20.7	24.5	27.0	29.2	32.9	33.1	32.7	31.8	28.2	22.8	19.7
	2%	WB	12.0	14.8	17.3	20.0	23.5	25.8	26.5	26.1	24.7	21.9	17.1	14.0
		MCDB	15.7	18.6	21.6	24.9	28.4	31.9	32.8	32.8	30.8	25.8	20.5	16.5
	5%	WB	9.2	12.2	15.8	18.9	22.5	25.0	25.8	25.3	23.9	20.5	15.5	11.4
		MCDB	13.4	15.8	19.8	23.4	27.3	30.9	32.2	32.2	29.7	23.8	19.0	13.4
	10%	WB	6.8	9.1	14.1	17.6	21.6	24.2	25.3	24.5	22.9	18.9	13.7	8.5
		MCDB	10.7	13.7	18.1	22.0	26.0	29.5	31.8	31.2	28.1	22.6	17.1	11.4
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.9	12.3	12.0	11.2	10.9	11.3	12.1	12.5	12.8	12.3	11.1
		MCDBR	16.4	16.1	15.2	14.1	12.5	12.0	13.4	14.6	13.9	14.6	14.8	14.8
		MCWBR	10.9	10.7	8.9	7.8	6.1	5.2	4.3	4.5	5.7	7.4	9.0	10.3
	5% WB	MCDBR	14.3	14.3	11.8	11.4	10.5	10.8	11.0	11.8	11.7	10.7	12.3	11.9
		MCWBR	10.3	10.6	8.1	7.6	5.9	5.3	4.7	5.0	5.2	6.6	8.8	10.0
Clear-Sky Solar Irradiance	taub	0.298	0.310	0.349	0.382	0.418	0.424	0.438	0.442	0.387	0.346	0.310	0.290	
	taud	2.472	2.452	2.382	2.318	2.275	2.283	2.290	2.255	2.374	2.467	2.509	2.513	
	Ebn at Noon	894	926	914	898	867	859	843	831	863	872	873	881	
	Edh at Noon	82	94	112	126	134	133	131	133	111	92	78	73	
All-Sky Solar Radiation	RadAvg	2.42	3.08	4.05	5.12	5.71	6.56	6.54	5.87	4.91	3.61	2.60	2.06	
	RadStd	0.23	0.49	0.25	0.34	0.53	0.57	0.43	0.37	0.38	0.43	0.29	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
(46)	Station Only	N/A	N/A	N/A	-1.25	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page